

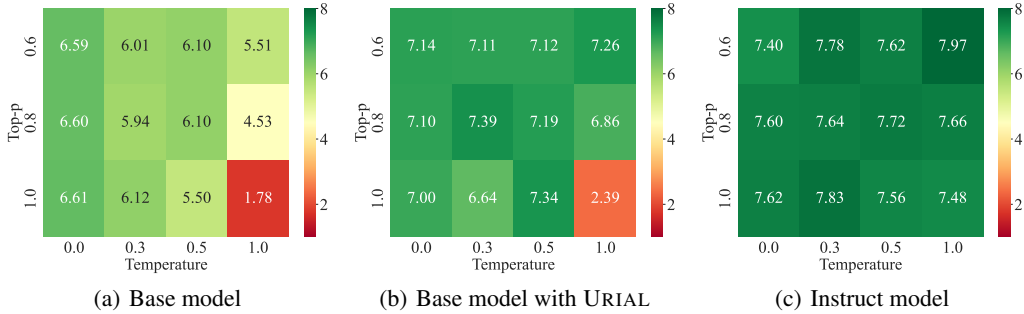


Figure 1: **Effect of decoding parameters on the 1st-turn MT-Bench scores across models.** We vary temperature and top- p , with a fixed repetition penalty of 1.15 as used in the URIAL codebase. The heatmaps show that the answering quality of the base model (Mistral-7B-v0.2) with and without URIAL is sensitive to the decoding schemes. Conversely, the performance of the instruct model is robust to varying the decoding parameters. Surprisingly, with proper decoding parameters, the base model alone is already capable of following instructions. See the complete results in App. E.3.

2.2 WHAT MATTERS FOR IN-CONTEXT ALIGNMENT VIA URIAL?

Decoding parameters. When computing the performance of URIAL, Lin et al. (2024) use decoding parameters (temperature = 0, top- p = 1, repetition penalty = 1.15) which differ from the default ones in MT-Bench (where temperature depends on the topic of the question, top- p = 1, repetition penalty = 1). To clarify the influence of these parameters that were not explicitly discussed in Lin et al. (2024), we compute the performance of base LLM, base LLM with URIAL, and fine-tuned LLM when varying decoding configurations. In Fig. 1 we fix the repetition penalty to 1.15, and vary temperature and top- p , for the Mistral-7B-v0.2 models. Surprisingly, we notice that with the decoding parameters of URIAL (temperature = 0, top- p = 1) even the base model without any in-context example achieves reasonable MT-Bench score (6.61 vs 7.00 of URIAL). Moreover, the temperature value appears to have the most influence of the performance of the base model. With URIAL, almost all configurations provide very similar results, with the exception of temperature = 1, top- p = 1 (possibly because it allows the sampling of generated tokens from the tail of the token distribution, thus producing low-quality text). Finally, the aligned model performs similarly across all decoding schemes, with slightly better results than URIAL. These results suggest that (1) the decoding parameters are an *overlooked factor* contributing to the success of in-context alignment, and (2) fine-tuning adjusts the sampling distribution of language models so that even with high-variance decoding configurations the generated text preserves high quality. We provide additional analysis for no repetition penalty (i.e. = 1) in Fig. 13 and Llama-3.1-8B in Fig. 14 in App. E.3: these extensive experiments further validate, with different base LLMs and settings, that the decoding scheme is a crucial factor for the instruction-following behavior of base models, and also affects, but to lower degree, in-context alignment. Instead, the fine-tuned models are robust to variations in the decoding parameters. Then, for the rest of the experiments we fix the decoding scheme of URIAL.

In-context prompt. Next, we want to disentangle the influence of the various elements of URIAL on instruction following performance. In Fig. 2 we track the MT-Bench scores when putting in-context all possible subsets of the three demonstrations of URIAL (i.e. from 0 to 3 examples are used). Moreover, for each case we test either adding or not the part of the prompt including the rules to follow. First, we find that neither 1st-turn nor 2nd-turn MT-Bench scores are significantly influenced by the addition of the set of rules, in turn highlighting the importance of the examples. Focusing on 1st-turn score (left plot in Fig. 2), we see that increasing the number of in-context demonstration progressively improves the performance of the base model (Mistral-7B-v0.2 in this case): this observation motivates us to add further high-quality in-context demonstrations, see Sec. 3. Finally, the 2nd-turn has the opposite trend, getting degraded by more (single-round) demonstrations.

2.3 IMPORTANCE OF QUESTION-ANSWER MATCHING

Surprisingly, Min et al. (2022) showed that using demonstrations with *random labels* does not significantly impair the results of ICL on classification and multiple choice tasks. We conduct a similar study for instruction following, see Table 2. Denoting $\{(X_i, Y_i)\}_i$ the set of in-context examples,

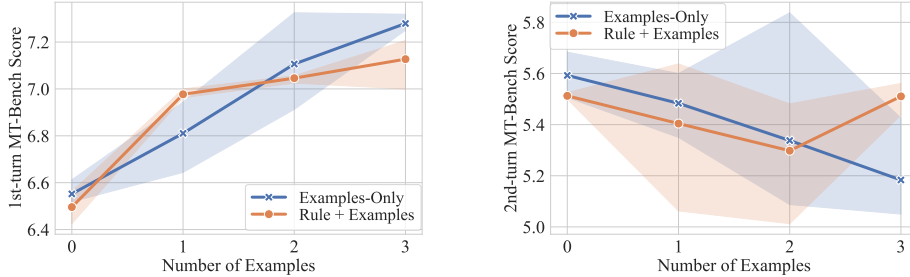


Figure 2: **Influence of the individual components of URIAL.** We report 1st and 2nd turn MT-Bench score for every subset of the three demonstrations of URIAL with Mistral-7B-v0.2. We test each configuration with (“Rule + Examples”) and without (“Examples-Only”) the URIAL set of rules in the in-context prompt. We observe a clear increasing trend in 1st-turn score with more examples, but a decrease in the 2nd-turn performance. The set of rules does not seem to influence the results. The randomness for 0 and 3 examples is caused by the small fluctuations in the score of the GPT-4 judge.

Table 2: **Importance of question-answer matching of demonstrations for in-context alignment.** We report the 1st-turn MT-Bench score of Mistral-7B-v0.2 and Llama-3.1-8B when varying the structure of the in-context examples $\{(X_i, Y_i)\}_i$, where X_i is the query and Y_i is the corresponding ground-truth answer.

In-context prompt	URIAL (3 examples)		URIAL + Greedy Search (6 examples)	
	Mistral-7B-v0.2	Llama-3.1-8B	Mistral-7B-v0.2	Llama-3.1-8B
no demonstrations	6.52	6.25	6.52	6.25
X only (no Y)	5.90	4.48	5.53	5.10
Y only (no X)	<u>6.94</u>	<u>6.83</u>	<u>7.02</u>	<u>6.98</u>
circular shift of Y	5.04	5.69	5.59	2.13
in-domain random Y	6.24	6.26	4.63	5.49
out-of-domain random Y	4.13	3.73	1.50	4.36
correct Y	6.99	6.95	7.43	7.81

with X_i the query and Y_i the corresponding ground-truth answer, we test several configurations on two sets of $\{(X_i, Y_i)\}_i$: the 3 URIAL examples and the 3 URIAL examples with the 3 examples found by our greedy search from Sec. 3.2 (to check the effect of increasing the number of demonstrations).

First, we do not use any demonstration, i.e., the original base models: with the decoding parameters found in the previous section, this already achieves reasonable results. Surprisingly, providing *the questions without the answers* (second row in Table 2) degrades the performance, while the opposite, *answers only*, is effective (0.4-0.7 higher scores than the base model). Next, we permute the answers Y_i s with a circular shift of one position, so that all correct answers are still contained in the prompt but matched with the wrong question: this significantly degrades the performance, especially with more examples (URIAL + Greedy Search), e.g. Llama-3.1-8B attains a score of only 2.13 (note that the minimum score is 1). Also, for each question, we sample a new answer from those provided for other instructions in the same category (*in-domain*): e.g., a question about coding is paired with an answer from a different coding question, ensuring that, while the content may be incorrect, the style remains appropriate. Although worse than the original one, this configuration achieves decent scores. Finally, we sample answers from instructions belonging to different *out-of-domain* categories, so that even their style does not match what expected for each question: this leads to the worst performance.

These results show that not all the conclusions from Min et al. (2022) apply to ICL used for instruction following. Most importantly, using answers with correct content and, especially, correct style is crucial for the success of ICL. This property becomes even more evident when increasing the number of examples (to 6 instead of 3), and motivates us to use a highly-curated instruction dataset like SkillMix in experiments in the next sections. Finally, this result suggests that ICL for instruction following works differently compared to other less open-ended tasks.

2.4 MULTI-TURN ALIGNMENT VIA ICL

In IFT, Zhou et al. (2023) find that adding a few multi-turn examples to the training data significantly improves the model’s performance on multi-turn conversations. URIAL (Lin et al., 2024) shows that

Table 3: **In-context alignment with *multi-turn* examples.** Second-turn answers for all methods are generated using GPT-4o. Extending URIAL with two-turn examples improves the 2nd-turn MT-Bench score. This shows that in-context alignment is not limited to single-turn performance.

In-context prompt	Mistral-7B-v0.2			Llama-3.1-8B		
	Turn 1	Turn 2	Avg	Turn 1	Turn 2	Avg
Base model (no prompt)	6.52	5.68	6.10	6.25	5.17	5.71
Original URIAL (single-turn)	6.99	5.55	6.27	6.95	5.31	6.13
Two-turn URIAL	7.18	5.68	6.43	7.31	6.44	6.87
URIAL + 3 two-turn SkillMix examples (no separation tag)	7.46	5.88	6.67	7.21	6.43	6.82
URIAL + 3 two-turn SkillMix examples (separation tag)	7.20	6.31	6.76	7.22	6.71	6.97
URIAL + 6 two-turn SkillMix examples (no separation tag)	7.18	5.89	6.53	7.14	6.26	6.70
URIAL + 6 two-turn SkillMix examples (separation tag)	7.24	5.78	6.51	6.89	6.33	6.61

the base LLMs can handle multi-turn conversations even with only having single-turn demonstrations in the context, but it does not quantify the performance nor rigorously study the role of examples that have more than one turn of conversation. To bridge this research gap, we further test how the presence of multi-turn examples in the context influences the instruction-following performance of in-context aligned models.

Experimental setup. We experiment using two open source LLMs, Mistral-7B-v0.2 and Llama-3.1-8B, and evaluate on MT-Bench. To ensure the multi-turn examples we test have high quality, we select single-turn examples randomly from SkillMix as the basis, the first turn, and then prompt GPT-4o to ask a follow-up question after seeing the first-turn conversation and generate an answer by using an instruction "Here is a single-turn conversation between a user and an assistant, please ask a follow-up question and provide an answer.". We examine the influence of two-turn examples from two perspectives, the number of two-turn examples (3 vs 6), and the format, with or without separation tags like "Here are 3 single-turn/multi-turn examples:."

Results. In Table 3, we observe that extending 3 URIAL single-turn examples to 2-turn examples consistently improves the evaluation scores on both turns, showing that in-context alignment solely with multi-turn examples can take care of single-turn conversations. The presence of two-turn examples in the context always leads to improved 2nd-turn MT-Bench scores, for example, an increase from 5.31 to 6.43 after adding 3 two-turn SkillMix examples on top of URIAL on Llama-3.1-8B. Another finding from our experiments is that the format of placing demonstrations plays a pivotal role in multi-turn interactions. In particular, adding a separation tag to arrange demonstrations in two groups improves 2nd-turn scores, from 5.88 to 6.31 on Mistral-7B-v0.2 and from 6.43 to 6.71 on Llama-3.1-8B, although sometimes it hurts the 1st-turn performance. However, we find that increasing the number of two-turn examples in the context does not lead to more benefits with or without a better format.

3 MANY-SHOT IN-CONTEXT LEARNING FOR INSTRUCTION FOLLOWING

Sec. 2.1 indicate that URIAL alone is not able, in most cases, to reach the performance of instruct models. In the following, we explore whether we can close this gap. We focus on Mistral-7B-v0.2 (Jiang et al., 2023) and Llama-3.1-8B (Dubey et al., 2024) since (a) both URIAL and the aligned model achieve competitive performance in Table 1, and (b) these base models have context windows of 32k and 128k respectively, which can fit many examples (around 50 for Mistral-7B-v0.2 and more than 200 for Llama-3.1-8B). Details on the experimental setup are in App. A.

Data. We adopt the SkillMix (Kaur et al., 2024) dataset consisting 4,000 examples as the source of high-quality examples for our many-shot in-context alignment experiments. The query-answer paired data generation process of SkillMix explicitly combines diversity and difficulty through random skill combinations, where the set of core skills for instruction-following, such as developing training programs and digital proficiency, are extracted by prompting a powerful LLM like GPT-4-Turbo. We present more details of SkillMix, including some examples (Fig. 6 and Fig. 7), in App. A.1.